

Compression-Interleaved Parenthesization Defects in LLM Replay Artifact Graphs: A Predeclared Null Test

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July 14, 2026

Abstract

Low-rank compression can induce parenthesization sensitivity even though ordinary matrix multiplication is associative. We test whether this compression-interleaved defect is larger in LLM replay-artifact cycles than under an edge-spectrum-matched, graph-unconstrained orientation null. The source surface, eligible-cycle criteria, compression ranks, null procedure, aggregation hierarchy, and one-sided support rule were fixed before execution. Across twelve source runs and two nontrivial compression ranks, all 24 endpoints were informative and coverage-complete, and none met the predeclared criterion for directional excess. Descriptively, all 24 endpoint median log-ratios were negative; this reverse-direction regularity was not predeclared and is not treated as confirmatory evidence. Because the null preserves marginal edge spectra while breaking shared-node realizability, the test cannot determine whether the negative comparison reflects a genuinely smaller observed defect or inflation of the null caused by destroying shared-node coupling. The result therefore does not support the targeted excess claim under the frozen protocol and motivates a separate prospective test using graph-constrained nulls, held-out runs, and positive controls.

1 Introduction

Ordinary matrix multiplication is associative, but an approximation operator inserted between multiplications need not respect that identity. If Q_q denotes rank- q truncated singular-value reconstruction, then $Q_q(M_2M_1)M_0$ and $M_2Q_q(M_1M_0)$ can differ even though the uncompressed products are equal. The difference is a parenthesization defect created by the placement of compression, not a failure of matrix associativity.

Replay artifact graphs provide a concrete setting in which to measure this defect. Their nodes carry local low-dimensional representation frames, their directed edges carry overlap operators between frames, and their cycles expose whether locally composed transports agree around a shared graph structure. A larger defect on observed cycles than under the edge-spectrum-matched, graph-unconstrained orientation null would indicate that the frozen observable detects directional excess beyond marginal edge spectra. The present question is deliberately narrower than whether any defect is nonzero.

We report a predeclared test of that question over twelve source runs and two nontrivial compression ranks. Its three main empirical results are:

1. all 24 endpoints were informative and coverage-complete;
2. none of the 24 endpoints met the predeclared support rule for directional excess over the null; and
3. all 24 endpoint median log-ratios were negative, a complete-family direction reported descriptively because a reverse-direction hypothesis was not predeclared.

We refer to the frozen protocol internally as Gate12C-1. It reads the artifact surface created by the Gate12A structural-replay line and is distinct from the Gate12B observer-relative closure audit. That project history supplies the data and provenance, but the scientific subject of this paper is the compression-interleaved parenthesization test itself.

2 Related Work

Low-rank approximation and truncation order. Truncated singular-value reconstruction is the optimal fixed-rank matrix approximation in Frobenius norm under the classical Eckart–Young result [1]. That pointwise optimality does not imply that inserting truncation inside a product preserves associativity. More broadly, sequentially truncated multilinear decompositions treat the sequence of approximation operations as part of the algorithm and can differ from a single uninterleaved decomposition [2]. The present observable isolates the analogous order sensitivity for two specified intermediate matrix products.

Procrustes and polar transport. Orthogonal Procrustes alignment uses an SVD-derived orthogonal map to align two coordinate systems [3]; the orthogonal polar factor provides the closely related nearest-orthogonal construction [4]. These methods motivate treating overlap-derived orthogonal or partial-isometry maps as transports between local representation frames. Here the tested matrices are the frozen raw overlap operators, and the rank truncation acts on their two-edge products.

Discrete connections, gauges, and representation holonomy. Connection-Laplacian methods place orthogonal transformations on graph edges and use their composition to analyze high-dimensional data [5]. Gauge-equivariant neural networks formalize the demand that computations be independent of local frame choices [6]. Recent work on neural representation holonomy likewise composes Procrustes-based local alignments around loops and constructs a gauge-invariant statistic [7]. Our graph, source artifacts, null, and parenthesization observable are different, but the shared mathematical requirement is that a reported loop quantity not depend on arbitrary node-wise orthogonal bases.

Javidnia develops a discrete gauge-theoretic and sheaf-theoretic atlas of LLM representations, including spanning-tree gauge fixing and fundamental-cycle holonomy [8]. Our setting differs in both observational substrate and target quantity: we operate on frozen replay artifact graphs rather than context-chart representations, and test compression-interleaved parenthesization defects against a

predeclared edge-spectrum-matched, graph-unconstrained orientation null rather than estimating representational holonomy.

Pale-ale artifact studies. The source surface descends from Structural Replay under a frozen FP32 regime [9]. The transport-first note defines gauge-covariant closure telemetry on replay artifact graphs [10], and the Gate12B report studies bounded observer-relative closure signatures [11]. The current paper neither re-tests nor extends those empirical claims. It asks a separate, predeclared question about compression-interleaved defects relative to a specified null.

3 Formal Setup

Let $\mathcal{G} = (V, E)$ be a replay artifact graph. Each node v carries an orthonormal local frame U_v of common active rank r . A directed edge from v_i to v_j carries the raw overlap operator

$$C(v_j \leftarrow v_i) = U_{v_j}^\top U_{v_i},$$

which maps coordinates at v_i to coordinates at v_j . Matrices act on column vectors from the left, and products compose right-to-left.

For an ordered eligible triangle

$$v_0 \longrightarrow v_1 \longrightarrow v_2 \longrightarrow v_0,$$

write

$$M_0 = C(v_1 \leftarrow v_0), \quad M_1 = C(v_2 \leftarrow v_1), \quad M_2 = C(v_0 \leftarrow v_2).$$

The uncompressed cycle operator $M_2 M_1 M_0$ maps the v_0 coordinate space to itself.

If $X = U \Sigma V^\top$ has singular values in nonincreasing order, define the rank- q truncated reconstruction

$$Q_q(X) = U_q \Sigma_q V_q^\top.$$

For $\epsilon = 10^{-12}$, the boundary-gap diagnostic is

$$g_q(X) = \frac{\sigma_q(X) - \sigma_{q+1}(X)}{\max\{\sigma_1(X), \epsilon\}}.$$

The frozen protocol requires $g_q > 10^{-3}$ for both intermediate products, so the selected rank- q approximation is separated at its truncation boundary.

The two compression-interleaved parenthesizations are

$$L_q = Q_q(M_2 M_1) M_0, \quad R_q = M_2 Q_q(M_1 M_0),$$

and the scalar defect is

$$a_q = \|L_q - R_q\|_F.$$

Associator is used operationally for the defect induced by the frozen intermediate-compression rule; ordinary matrix multiplication itself remains associative.

Proposition 1 (Gauge invariance). *Assume*

$$\sigma_q(X) > \sigma_{q+1}(X) \quad \text{for } X \in \{M_2M_1, M_1M_0\}.$$

Then the scalar defect is invariant under independent node-wise orthogonal changes of basis. In particular, if $G_i \in O(r)$ and

$$C(v_j \leftarrow v_i)' = G_j^\top C(v_j \leftarrow v_i) G_i,$$

then $a'_q = a_q$.

Proof. The strict boundary separation makes the rank- q reconstructed matrix unique, although a particular choice of singular-vector basis within the retained or discarded subspace need not be unique. The reconstructed matrix therefore obeys the orthogonal covariance identity

$$Q_q(S^\top XT) = S^\top Q_q(X)T$$

for orthogonal S, T . For the ordered triangle,

$$M'_0 = G_1^\top M_0 G_0, \quad M'_1 = G_2^\top M_1 G_1, \quad M'_2 = G_0^\top M_2 G_2.$$

Cancellation of adjacent orthogonal factors and covariance of Q_q give

$$L'_q = G_0^\top L_q G_0, \quad R'_q = G_0^\top R_q G_0.$$

Thus $L'_q - R'_q = G_0^\top (L_q - R_q) G_0$. Orthogonal invariance of the Frobenius norm yields $a'_q = a_q$. $\square$

Proposition 1 makes a_q a well-defined scalar on the artifact graph rather than a diagnostic tied to one stored choice of local bases. Because the split-gap diagnostic depends only on singular values, eligibility is also invariant under node-wise orthogonal gauge changes. The edge-spectrum-matched, graph-unconstrained orientation null is defined from edge spectra, so the full observed-to-null comparison is gauge-independent under the frozen protocol. The runner’s deterministic gauge control evaluates a signed-permutation subset of these changes; Proposition 1 states the corresponding mathematical result for the full orthogonal class at strictly separated cuts.

4 Predeclared Experimental Design

The source surface comprised twelve already materialized replay-artifact runs: four model lines crossed with transcript, briefing, and archive rendering families. The three rendering families are distinct artifact constructions over the frozen source benchmark rather than independent task datasets. Table 1 gives the case map. Each eligible triangle was residual-bearing, had defined Gate12A holonomy, had three equal-rank-orthogonal legs with common rank $r = 3$, belonged to a valid single-sample source block, and had stable intermediate cuts. The two nontrivial ranks were $q \in \{1, 2\}$, evaluated at all three cyclic roots.

Exact model identifiers, in table order: Qwen/Qwen2.5-0.5B; Qwen/Qwen2.5-3B-Instruct; meta-llama/Llama-3.2-3B-Instruct; and Qwen/Qwen3-4B.

Table 1: Canonical source-run case map. Family labels give the number of predeclared source-sample blocks.

Model line	Transcript (128)	Briefing (200)	Archive (128)
Qwen2.5-0.5B	case_01	case_02	case_03
Qwen2.5-3B-Instruct	case_04	case_05	case_06
Llama-3.2-3B-Instruct	case_07	case_08	case_09
Qwen3-4B	case_10	case_11	case_12

The comparator was fixed as an *edge-spectrum-matched, graph-unconstrained orientation null*. For each observed edge spectrum Σ_e , null draw b used

$$\tilde{C}_e^{(b)} = O_{e,L}^{(b)} \Sigma_e O_{e,R}^{(b)\top},$$

with independently generated orthogonal left and right orientations. One randomized edge triple was reused across the three roots and both values of q within a cycle draw. The procedure requested 255 valid draws, allowed at most 1024 attempts when stable-cut requirements failed, and applied the same 10^{-3} split-gap threshold to observed and null products. The null preserves every edge’s singular values but is not required to arise from a single consistent choice of bases at shared nodes.

For one observed root and its valid null draws, the primary log-ratio was

$$d_q = \log \left(\frac{a_{q,\text{obs}} + \epsilon}{\text{median}_b a_q^{(b)} + \epsilon} \right).$$

The frozen hierarchy took medians over three roots to obtain a cycle/ q value, over valid cycles within each source-sample block to obtain a block score, and over represented blocks to obtain an endpoint median. Coverage required at least 90% of expected cycles and 90% of expected blocks, with no imputation.

The confirmatory test was one-sided in the positive direction. Block scores above 10^{-12} were positive, scores below -10^{-12} were negative, and intervening values were ties. If s of n non-tied block scores were positive, the frozen exact-binomial calibration was

$$p_{\text{raw}} = \sum_{k=s}^n \binom{n}{k} 2^{-n}.$$

Holm correction was applied across all 24 case/ q endpoints [12]. Endpoint support required complete coverage, an informative sign test, a positive endpoint median, and Holm-adjusted $p < 0.05$. This calibration is the predeclared decision rule; it is not a claim that source blocks form an independent population sample.

5 Results

None of the 24 predeclared endpoints supported directional excess. All endpoints were informative and coverage-complete. In addition, every endpoint median log-ratio was negative. The latter

complete-family direction was not a predeclared reverse-direction hypothesis and is reported descriptively only.

A post-hoc descriptive reporting addendum was committed before computing any new aggregate (commit 671e6d1). Under that boundary, exponentiating the 24 endpoint-level hierarchical summaries gives values from 0.028 to 0.236, with a median of 0.066 on that summary scale. These are transformed endpoint summaries, not raw-row median ratios or a reverse-direction test.

Figure 1 shows the result at a glance. Each plotted point is one predeclared endpoint, with the two ranks paired within each source run. The zero line is the boundary between an observed endpoint median above and below its corresponding null median; it is not a significance threshold.

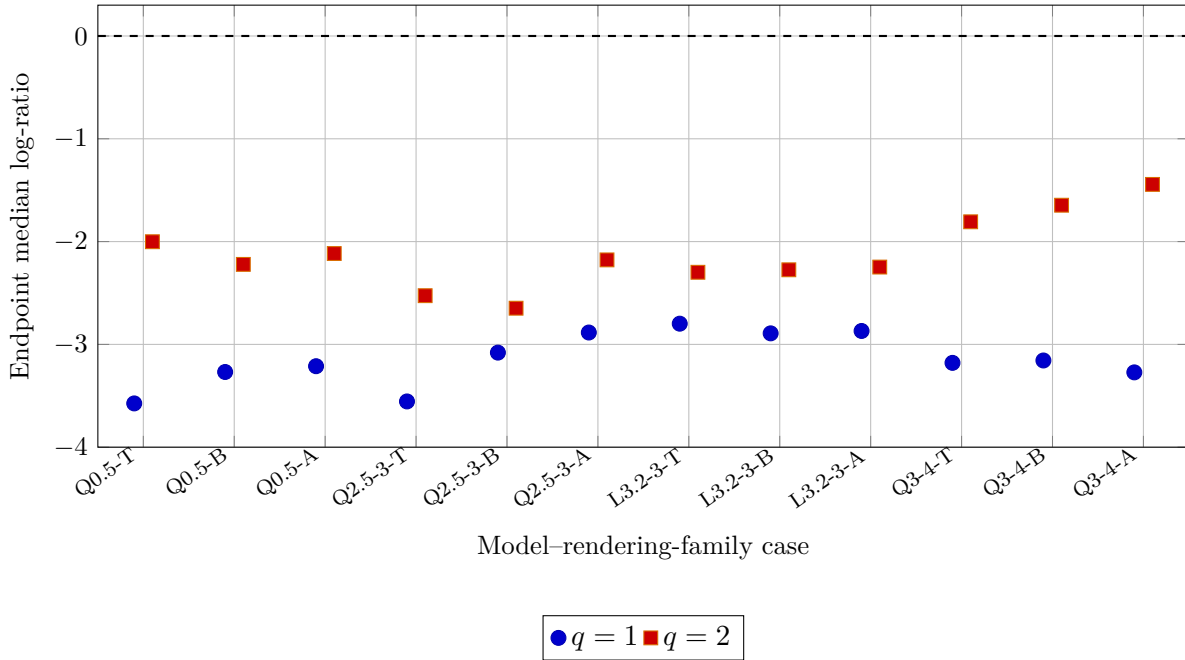


Figure 1: All 24 predeclared endpoint median log-ratios, offset horizontally by compression rank. T, B, and A denote transcript, briefing, and archive; Table 1 gives the full case map. All points fall below the zero line. This complete-family direction is descriptive, not a predeclared reverse-direction test.

Table 2 reports the complete primary endpoint family. The block sign counts are inputs to the one-sided positive-direction test. Every endpoint median remained negative and no endpoint met the frozen support rule.

Table 2: Complete primary endpoint results. The n_+/n_- column gives positive and negative non-tied source-block counts, not a ratio.

Case	Model	Family	q	Median	Blocks n_+/n_-	Support
C01	Q2.5-0.5B	Transcript	1	-3.574093	0/128	No
C01	Q2.5-0.5B	Transcript	2	-2.001354	0/128	No
C02	Q2.5-0.5B	Briefing	1	-3.268481	0/200	No
C02	Q2.5-0.5B	Briefing	2	-2.221682	0/200	No
C03	Q2.5-0.5B	Archive	1	-3.212260	0/128	No
C03	Q2.5-0.5B	Archive	2	-2.116187	7/121	No
C04	Q2.5-3B-I	Transcript	1	-3.554964	0/128	No

Table 2 continued

Case	Model	Family	q	Median	Blocks n_+/n_-	Support
C04	Q2.5-3B-I	Transcript	2	-2.526256	0/128	No
C05	Q2.5-3B-I	Briefing	1	-3.080008	0/200	No
C05	Q2.5-3B-I	Briefing	2	-2.647950	0/200	No
C06	Q2.5-3B-I	Archive	1	-2.884508	0/128	No
C06	Q2.5-3B-I	Archive	2	-2.177989	0/128	No
C07	Llama-3.2-3B-I	Transcript	1	-2.798490	0/128	No
C07	Llama-3.2-3B-I	Transcript	2	-2.298432	0/128	No
C08	Llama-3.2-3B-I	Briefing	1	-2.892626	0/200	No
C08	Llama-3.2-3B-I	Briefing	2	-2.274441	0/200	No
C09	Llama-3.2-3B-I	Archive	1	-2.869399	0/128	No
C09	Llama-3.2-3B-I	Archive	2	-2.247802	0/128	No
C10	Qwen3-4B	Transcript	1	-3.179593	0/128	No
C10	Qwen3-4B	Transcript	2	-1.807492	1/127	No
C11	Qwen3-4B	Briefing	1	-3.156542	0/200	No
C11	Qwen3-4B	Briefing	2	-1.646118	9/191	No
C12	Qwen3-4B	Archive	1	-3.272157	0/128	No
C12	Qwen3-4B	Archive	2	-1.443657	0/128	No

The confirmatory conclusion follows from the frozen positive-direction rule: 0/24 endpoints supported excess over the edge-spectrum-matched, graph-unconstrained orientation null. This was not a coverage-limited outcome; all 24 endpoints represented every expected cycle and source block and had an informative sign test. Appendix A gives the recorded raw and Holm-adjusted values.

The descriptive conclusion is separate. The endpoint family itself was fixed before execution, so its full direction pattern can be reported without selecting a favorable subgroup. Nevertheless, the hypothesis that all or most endpoints would be negative was not predeclared, and no reverse-direction test, subgroup analysis, or additional significance claim is introduced here.

6 Discussion

6.1 What the predeclared test rules out

The result blocks the targeted claim that replay-artifact cycles show positive compression-interleaved defect excess over this null under the frozen ranks, coverage criteria, aggregation hierarchy, and support rule. It does not show that the observed defect is identically zero, and it does not establish a general absence of graph structure.

Gate12B addresses a different question: a bounded observer-relative closure signature on an existing artifact surface. Gate12C-1 supplies neither positive evidence for that result nor a refutation of it. The connection is procedural: the same research program can report a bounded positive artifact result while declining a stronger, unsupported excess claim.

6.2 Descriptive reverse-direction regularity

All 24 endpoint medians were negative, spanning every model line, rendering family, and tested rank in the predeclared grid. This is a conspicuous complete-family pattern and belongs in the

result record. It remains descriptive because the protocol did not predeclare a reverse-direction hypothesis or a decision rule for family-wide negativity. We therefore do not label it confirmatory evidence, a suppression effect, or a discovery of graph coherence.

6.3 What the current null cannot distinguish

The null matches each edge’s singular spectrum but samples its left and right orientations independently. At a shared node, adjacent null edges therefore need not use one common latent basis. The null can generate edge triples that are not realizable by any single node-wise assignment of frames, even though each edge is marginally spectrum-matched.

Consequently, the negative endpoint medians admit more than one explanation. They may reflect an absence of the targeted positive excess, an increase in null defects caused by destroying shared-node coupling, or a combination of the two. The present design does not identify which explanation is operative. It therefore supports the observed directional comparison but not the causal claims that real graphs are coherent or that graph consistency produced the negative direction.

6.4 Next prospective test

A separate Gate12C-2 plan should predeclare the reverse-direction hypothesis and evaluate it on held-out runs. Its null should be graph-constrained and node-consistent, so all incident edges inherit compatible orientations from shared node transformations. A synthetic positive control should establish that the pipeline detects a known compression-interleaved effect, and frozen sensitivity checks should test dependence on rank, split-gap threshold, and null construction. None of those prospective results is imported into the current paper.

7 Threats to Validity

- **Edge-spectrum-matched, graph-unconstrained orientation null.** Spectrum matching is edge-wise; shared-node realizability is not preserved. This is the central interpretive limitation of the comparison.
- **No predeclared positive control.** The execution verified algebraic, gauge, coverage, and deterministic-output controls, but did not include a synthetic instance with a known positive directional effect. Sensitivity to such an effect remains to be demonstrated prospectively.
- **Fixed source surface.** The evidence is limited to twelve source runs from four dense-transformer model lines and three rendering families. It is not a population-wide claim about LLMs.
- **Decision calibration.** Source-sample blocks avoid treating overlapping roots and cycles as separate observations, but independence or exchangeability across blocks is not established. The exact-binomial calculation is therefore read as a frozen decision calibration, not broad population inference.

- **Equal-rank construction.** The common node rank was three and the tested compression ranks were one and two. Rectangular rank mismatch and other compression rules remain outside the claim.
- **Artifact-level scope.** The observable is defined on replay artifacts. It is not a model-quality ranking, an answer-correctness classifier, a safety guarantee, or a weight-level causal mechanism.

8 Data and Code Availability

The protocol, code, and tracked endpoint record are available in the public pale-ale repository. The reproducibility surface is indexed as follows:

- the pre-execution protocol and canonical case map are in `workstream/235_GATE12C1_FIRST_EMPIRICAL_EXECUTION_PLAN.md`;
- the complete endpoint summary and provenance digests are in `workstream/236_GATE12C1_FIRST_EMPIRICAL_RESULT_MEMO.md`;
- the frozen post-hoc descriptive reporting boundary is in `workstream/240_GATE12C1_POST_HOC_DESCRIPTIVE_REPORTING_ADDENDUM.md`;
- the measurement runner is `tools/run_gate12c_compressed_overlap_associator.py`.
Frozen commit: 8d5613bffe5b6c91d0956c812404072eb76e98c6.
SHA256: b363fd874a0538dc548853e97e8ec17c0eb84be5658f6e2f01f60d2a12789c3e;
- the grid summarizer is `tools/summarize_gate12c1_first_empirical_grid.py`.
Frozen commit: 6b66f49710f4b7cbdae7e5a282d14d1d81723b30.
SHA256: 9e527c0363875de912f9abf9ba38acb462a13b0b62da6e14be35d4c3526df708;
- the executed case manifest has SHA256
dae1f2690fc707a166277dac143aafcfa5e51e23934428861ddc48cac97ad704.

The companion archival capsule is available at doi:10.5281/zenodo.21355572. The capsule contains the immutable `case_manifest.json`, per-case runner manifests, and their checksum receipts. It also contains `gate12c1_run_q_tests.csv`, the cycle- and block-level score exports, the summarizer manifest and checksums, the software-environment receipt, and `CHECKSUMS-SHA256.txt`.

The corresponding repository snapshot is identified by this immutable tag:

`gate12c1-parenthesization-defects-v1.0.0`

9 Conclusion

We tested whether compression-interleaved parenthesization defects in LLM replay-artifact cycles exceeded an edge-spectrum-matched, graph-unconstrained orientation null. All 24 predeclared end-

points were informative and coverage-complete, and none supported the targeted positive excess claim. Descriptively, all 24 endpoint median log-ratios were negative. That reverse-direction regularity was not predeclared and is not confirmatory evidence. Because the null breaks shared-node realizability, the current test cannot identify whether graph-level coupling contributed to the direction of the comparison. A prospective held-out test with a node-consistent null, reverse-direction rule, positive control, and frozen sensitivity analysis is required to answer that question.

A Endpoint Directional-Test Audit

Table 3 reports the frozen one-sided positive-direction test outputs. Values are shown as recorded by the summarizer after conversion to double precision. For endpoints with a nonzero positive-block count, a displayed raw value of 1.0 can represent an exact upper-tail probability below one but within machine precision of it; Holm-adjusted values are capped at one.

Table 3: Recorded endpoint directional-test values and support calls.

Case	q	Recorded raw p	Holm p	Support
C01	1	1.0	1.0	No
C01	2	1.0	1.0	No
C02	1	1.0	1.0	No
C02	2	1.0	1.0	No
C03	1	1.0	1.0	No
C03	2	1.0	1.0	No
C04	1	1.0	1.0	No
C04	2	1.0	1.0	No
C05	1	1.0	1.0	No
C05	2	1.0	1.0	No
C06	1	1.0	1.0	No
C06	2	1.0	1.0	No
C07	1	1.0	1.0	No
C07	2	1.0	1.0	No
C08	1	1.0	1.0	No
C08	2	1.0	1.0	No
C09	1	1.0	1.0	No
C09	2	1.0	1.0	No
C10	1	1.0	1.0	No
C10	2	1.0	1.0	No
C11	1	1.0	1.0	No
C11	2	1.0	1.0	No
C12	1	1.0	1.0	No
C12	2	1.0	1.0	No

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